Lowell Welburn

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Professional Summary

A Ph.D. candidate in Mechanical Engineering with a foundational background in Biomechanics/Kinesiology specializing in robotics, automation, and bio-inspired design. Possesses hands-on experience designing and fabricating novel robotic systems using SolidWorks and 3D printing, developing reinforcement learning control strategies, and creating physics-based simulations in MuJoCo and MATLAB. Seeking to advance research in bio-inspired robotics and intelligent control systems, while mentoring others in engineering principles and biomechanics.

Education

Doctor of Philosophy in Mechanical Engineering — **Expected Graduation August 2026**
North Carolina A&T State University — Greensboro, North Carolina

- Focus on Robotics and Automation (3.64 GPA)

Master of Science in Mechanical Engineering — **Graduation August 2023**
North Carolina A&T State University — Greensboro, North Carolina

- Focus on System Control and Biomedical Engineering (3.48 GPA)

Bachelors of Science in Exercise Physiology — **Graduation July 2018**
Brigham Young University - Idaho — Rexburg, Idaho

- Emphasis in Movement and Rehabilitation Science (3.92 GPA, full-tuition academic scholarship)

Experience

Robotic Hand Research Assistant — **January 2025-Present**
North Carolina A&T State University — Greensboro, North Carolina

- Designed novel bioinspired robotic hand using SolidWorks CAD and fabricated with 3D printing
- Transferred CAD to MuJoCo for physics-based simulation and reinforcement learning training before transitioning to ROS 2 while employing OpenCV in Python to run computer vision 6 DoF pose estimation
- Created safe lab environment for and worked closely with commercial robots such as Agility Robotics Digit, Boston Dynamics Spot, and the Quanser QArm

Intern, Minority Serving Institutions Internship Program (MSIIP) — **June 2025-Present**
Oak Ridge National Laboratory — Oak Ridge, Tennessee

- Trained reinforcement learning real-time decision-making for hybrid zero-emission vehicle stations
- Evaluated the performance of machine learning compared with traditional and manual techniques

Biomechanics Lab Instructor — **January 2026-May 2026**
North Carolina A&T State University — Greensboro, North Carolina

- Mentored minority students as an interdisciplinary instructor from a unique educational background

NASA OSTEM Intern — **August 2023-August 2024**
NASA Glenn Research Center — Greensboro, North Carolina

- Created MATLAB-based 1D numerical model for simulating two-phase cryogenic line chilldown
- Presented simulation results for various test conditions, including microgravity vs terrestrial gravity

Hypersonic Research Assistant **May 2022-August 2023**
North Carolina A&T State University Greensboro, North Carolina
- Investigated traditional vs RL control methods for hypersonic vehicle actuators within MATLAB/Simulink
- Manufactured a test setup including a linear servo, LVDTs, steppers, and custom spring mechanisms

Assistant Center Director **March 2020-August 2020**
Mathnasium Boise, Idaho
- Organized coursework for 70+ students, while coordinating employee schedules to meet demands
- Instructed multiple K-12 students simultaneously in a wide range of Math subjects

Anatomy/Exercise Physiology Lab Instructor **September 2018-April 2019**
Brigham Young University - Idaho Rexburg, Idaho
- Managed coursework of over 130 students each semester
- Lectured in front of 30+ classroom sizes
- Provided individual feedback to students

Anatomy/Kinesiology Tutor **September 2016-April 2018**
Brigham Young University - Idaho Rexburg, Idaho
- Instructed and tutored Anatomy and Physiology students for more than 300 teaching hours

Publications

- ReAct Hand: Coactivation-Aware Control for Bioinspired Remote-Actuated Robotics, 2026, CoRL (In-Progress)
- Reinforcement Learning-Based Real-time Decision Making for Zero-Emission-Vehicle Stations, 2026, NAPS (In-Progress)
- Prospects and Trends in Biomedical MEMS Devices: A Review, 2025, *Biomolecules*
- Comparative Study of a DC Stepper Motor Linear Actuator with a Hybrid Piezo-hydraulic Actuator for a Control Surface of a Glide-body Vehicle, 2024, *JANNAF*
- Dynamics and Adaptive Control of a Hybrid Piezoelectric-hydraulic Actuator for Hypersonic Morphing Control Surface, 2023, *JANNAF*
- Control of a Piezoelectric Actuator for Hypersonic Application Using Feedforward Compensation and a PID Controller Tuned by Reinforcement Learning, 2023, *NCAT Masters Thesis*

Achievements

- Aerospace UPLIFT Seminar Speaker February 2026
- Engineer In Training (EIT) April 2025
- Graduate Research Poster Competition Winner April 2025
- SolidWorks CSWP March 2019
- Convocation Speaker July 2018

Skills

- Python, MATLAB, Simulink, SolidWorks, Sim2Real, Linux, ROS 2, OpenCV, MuJoCo, Reinforcement Learning, Machining, 3D Printing, Microcontroller, Presentation, Teaching

Volunteer

Missionary for The Church of Jesus Christ of Latter-day Saints **July 2013-July 2015**
Oklahoma, Tulsa Mission Oklahoma, Arkansas, Missouri, Kansas
- Communicated with the deaf, visually impaired, and those of diverse nationalities and backgrounds
- Worked 16 hour days for 2 consecutive years managing time, money, and resources
- Motivated others by serving in leadership positions such as District Leader, Trainer, and Zone Leader